

A STUDY OF INDIVIDUAL BOUNDARY-SPANNING COMMUNICATION PATTERNS IN A RESEARCH AND DEVELOPMENT SETTING

Ann C. Séror

Département de Management, Université Laval, Ste-Foy, P.Q., G1K 7P4 (Canada)

ABSTRACT

This study explores patterns of individual boundary-spanning communication in a government-managed research and development setting. Individual position in the organization

hierarchy and preference for risk, change, and new experiences are the most significant antecedents of general levels of boundary-spanning communication.

INTRODUCTION AND RESEARCH OBJECTIVES

A call has been made to develop new models of organizational systems which include characteristics of both individuals and situations as determinants of individual behavior in organizations (Campbell et al., 1970; James and Jones, 1976). The development of such integrated approaches requires analysis of both macro and micro dimensions of organizations for an understanding of individual behavior and attitudes in organizational settings. The synthesis requires (James and Jones, 1976): (1) the elaboration of models including total organizations, their subsystems and socio-cultural and economic environments; (2) the identification of dimensions of organizations which may be relevant across a variety of service as well as industrial systems in both the public and private sectors; and (3) the study of dynamic organizational processes.

Some recent research has focused on processes of communication across organizational boundaries (House and Rizzo, 1972; Keller and Holland, 1975; Leifer and Huber, 1977; Bacharach and Aiken, 1977; Tushman, 1977, 1979a, b). This research has suggested the importance of the individual perception of task uncertainty as an antecedent of communication across organizational boundaries. Perceived task uncertainty has been defined and operationalized in a variety of ways: ambiguity and conflict (House and Rizzo, 1972; Keller and Holland, 1975), task predictability (Leifer and Huber, 1977), and task variability (Bacharach and Aiken, 1977; Tushman, 1977, 1979a, b). Keller and Holland (1975) found positive but nonsignificant associations between level of individual boundary-spanning activity and perceived role conflict and ambiguity. Tushman (1979a, b) found support for the hypothesis that under the condition of perceived environmental variability or uncertainty, the difference in work unit levels of communication between more and less rou-

tine task areas would be greater than under more stable environmental conditions. Leifer and Huber (1977) found positive and significant associations between perceived environmental uncertainty, organicness of structure, and level of boundary-spanning and organicness of structure (Burns and Stalker, 1961) were partialled out, respectively. Similarly Bacharach and Aiken (1977) found that perceived task routineness did not emerge as a significant antecedent of vertical and horizontal communications or boundary-spanning communication. However, they reported significant negative zero-order correlations between perceived task routineness and dimensions of communication for department heads. (This association was not significant for subordinates.) Bacharach and Aiken recommended further research on communication patterns at different levels of organization hierarchy.

According to Huber et al. (1975), the increased attention to the perceived uncertainty construct is a response to the need to apply organization theory to practice. They formulate the proposition that organizational structures, processes, and effectiveness are affected by perceptions of uncertainty and other dimensions of the task environment. There is a need to demonstrate the perception of uncertainty as it is affected by variables under the control of organization designers. It is the intent of this study to investigate patterns of individual boundary-spanning communication and perceived task uncertainty in the field setting of six research and development laboratories under contract to the United States Department of energy and managed under the electric and Hybrid Vehicle Development (EHV) Program in 1980. The following research questions are under consideration:

(1) How are individual differences in preference for risk, change, and new experience related to perceptions of task uncertainty, other task environmental characteristics, and levels of boundary-spanning communication?

(2) What task-environmental characteristics are associated with perceived task uncer-

tainty and individual boundary-spanning communications?

(3) How are individual length of tenure with the EHV Program and level of hierarchy in project organization related to perception of task uncertainty and level of boundary-spanning communication?

(4) How does perceived task uncertainty mediate the associations between task-environmental characteristics and level of boundary-spanning communication?

The answers formulated to these questions will suggest recommendations for future research efforts.

METHOD

Setting and sample

This study was conducted in Spring, 1980, in the setting of the Electric and Hybrid Vehicle Development Program administered by the United States Department of Energy. The Program was initiated by Congress with passage of the Electric and Hybrid Vehicle Research, Development, and Demonstration Act of 1976 (Public Law 94-413). The Act was passed to foster commercialization of electric vehicles with two broad objectives under consideration: (1) to reduce national oil consumption and (2) to achieve environmental benefits. The Act first assigned administrative responsibility to the Energy Research and Development Administration which was later restructured to create the Department of Energy. Funding was provided for a program of research and development, vehicle demonstrations, and incentives to private production of electric and hybrid vehicles, with provisions to protect and encourage the participation of small business. The program was intended to launch a new technology and to be terminated when the new technology became viable in the free market place and private industry took over its promotion. Since its inception in 1976, the EHV Program had ma-

tured. In terms of the range of Program tasks, the emphasis had extended from the pursuit of basic research to the development of plans for commercialization and the development of new markets.

Not only had the range of tasks encompassed by the Program grown, but Program management had developed. At the outset in 1976, there was a high degree of ambiguity in task definition, and the research and development laboratories seeking contracts under the EHV Program were, to a considerable extent, in competition for funding. As the program grew, however, the complementary roles and contributions of the research and development organizations under contract to the Department of energy in pursuit of EHV Program goals were better defined, and communication among all Program participants became more open. The opening of communication had a major consequence in the reduced emphasis on formal coordination mechanisms and the increased need for informal exchange of task-related information.

The field for this study was the network of organizations under contract to the Department of Energy to carry out the research, development, and demonstration of the electric vehicle. The organizations included in the study were NASA Lewis Research Center, Jet Propulsion Laboratory, Aerospace Corporation, Booz-Allen and Hamilton, Argonne National Laboratory, and Lawrence Livermore Laboratory.

The sample for this study included the management staff for the project task structures. The sampling objective was to include all those responsible for managing the tasks in progress within each project. The criteria for selection of subjects for interview were:

(1) Involvement in the EHV Program at the level of project planning and management for at least six months.

(2) Responsibility for tasks concerned with development of some technical aspect of the electric or hybrid vehicle or its commercialization.

The final sample consisted of 52 male professionals in the management of six projects, with six to eleven interviewees reporting from each project.

Data collection procedure

The data were collected by individually administered questionnaire interviews conducted at the offices of each of the research and development laboratories. The interview format was developed through a series of discussions with management of the EHV Program, including the Chief of the Technology and Systems Development Branch and the Resource Manager of the Office of Commercialization, in February 1980. Six preliminary interviews were conducted to inform Program managers of the type of data to be collected and to give them an opportunity to make suggestions for revision of the format. As a result of these discussions, the final interview format was determined. Program managers agreed to notify project managers in the six field organizations of forthcoming visits in April and May 1980, for data collection. Each project manager was responsible for setting a two-day agenda for individual interviews. In addition, information was requested from each project office describing the overall organization and tasks in progress.

Measures

The interview format included measures of the following variables (see Appendix A):

1. Task routinization

The research tasks being performed in each project organization were categorized on a continuum to reflect activities of an increasingly routine nature. Tushman (1977) developed the following categorization of research activities: (a) *Basic research* is inquiry of a general nature for a broad range of applications or for the development of new knowledge about an area. (b) *Applied research* is work involving basic knowledge for creation and evaluation of new con-

cepts. (c) *Development* is the combination of existing concepts with new knowledge to provide a new product or process. (d) *Technical service* is cost/performance improvement of existing products, processes or systems; or recombination, modification and testing of systems using existing knowledge. (e) *Commercialization* includes activities with the objective of opening new markets for existing products. Task routinization was calculated as the weighted sum of the percentages of work reported in each category by interviewees. Commercialization is the most routine research category, and basic research is the least routine.

2. Planning activity

Planning activity is defined as the extent to which a task is effectively structured and important factors taken into consideration before the task is started (House and Rizzo, 1972). This scale is composed of five statements requiring ratings of the extent to which each describes the subject's task environment.

3. Sensation-seeking tendency

This individual characteristic is defined as the preference for a high stimulus environment and may be considered closely related to the preferred arousal level. This characteristic is measured on a scale composed of 17 items selected from an instrument developed by Mehrabian and Russel (1974).

4. Task interdependence

Task interdependence is defined as the extent to which the accomplishment of a task requires interrelationship with task areas in other research centers participating in the EHV Program. This variable is measured with one item (Tushman, 1977).

5. Perceived task uncertainty

This variable is defined on two dimensions: task variability and task difficulty (Van de Ven and Delbecq, 1974; Van de Ven et al., 1976; Van de Ven, 1977). Task variability is operationalized as the frequency of work exceptions en-

countered in a work unit, and task difficulty is the analyzability of work methods as suggested by Perrow (1967). The 10 items of the perceived task uncertainty measure were adapted from an instrument developed by Van de Ven (1977).

6. Boundary-spanning activity

Individual boundary-spanning activity is defined as the weekly frequency or extent of communication, formal and informal, written and spoken, across organizational boundaries between tasks within project organizations and between research centers. This self report instrument including eight items was developed by Leifer and Huber (1977). The items are used in this study to measure communication among task areas within projects (intra-project, BSA) and among research and development centers (center-to-center, BSB).

7. Tenure

Tenure with the new EHV Program was reported in months.

8. Hierarchy

Two levels of hierarchy were distinguished: project managers and their assistants (superior level), and section heads and task leaders (subordinate level).

In addition to the structured questionnaire, the interview format provided each interviewee with an opportunity to comment on the history of the EHV Program and the effectiveness of its management by the Department of Energy.

RESULTS AND DISCUSSION

Table 1 shows the properties of measures used in this study. The research questions to be discussed are the following:

TABLE 1

Properties of measures

Measure (c)	Number of items	Mean	Standard deviation	Reliability
1. Task routinization by classification (TR)	1	329.808	75.355	-
2. Planning activity (PA)	5	4.312	1.097	0.641 ^a
3. Sensation-seeking tendency (SS)	17	5.855	0.935	0.796 ^a
4. Task interdependence (TI) center-to-center	1	5.269	2.978	0.277 ^b
5. Perceived task uncertainty (PTU)	10	6.67	1.099	0.740 ^a
6. Boundary-spanning activity (BS)	8	5.477	3.308	0.760 ^a
a. Intra-project (BSA)	4	6.014	3.992	0.791 ^a
b. Center-to-center (BSB)	4	4.952	3.348	0.659 ^a
7. Tenure with the EHV program (months) (T)	1	31.115	17.333	-
8. Hierarchy (H)	1	1.288	0.457	-

^aCoefficient alpha.^bTest-retest reliability ($n=26$).^c $n=52$.

1. How are individual differences in preference for risk, change, and new experience related to perception of task uncertainty, task-environmental characteristics, and individual level of boundary-spanning communication?

Correlational analyses in Table 2 show that individual sensation-seeking tendency is significantly associated with perceived task uncertainty ($r=0.330$, $p\leq 0.05$), task routinization by classification ($r=-0.240$, $p\leq 0.10$) and with boundary-spanning activity ($r=0.250$, $p\leq 0.10$) although significance in the last two cases is marginal. The greater the individual sensation-seeking tendency, the greater the perception of task uncertainty, and the greater the level of boundary-spanning communication, particularly with other research centers. This finding is consistent with the proposition offered by Mehrabian and Russel (1974) that individuals high in sensation-seeking tendency will actively seek to increase their arousal by selecting novel, complex, or unpredictable settings.

2. What task-environmental characteristics are associated with perceived task uncertainty and individual level of boundary-spanning communication?

Center-to-center task interdependence is associated with the perception of uncertainty ($r=0.248$, $p\leq 0.10$). Perceived task uncertainty is associated only with center-to-center boundary-spanning communication ($r=0.231$, $p\leq 0.10$). For both of these relationships, significance is marginal. These results suggest that the greater the center-to-center task interdependence, the greater the individual perception of task uncertainty. The interdependence of the task may be a factor contributing to the perception of uncertainty as suggested by Galbraith (1977). Also, as the perception of task uncertainty increases, the level of individual boundary-spanning communication from center-to-center increases. This increased communication activity may be directed to information search for uncertainty reduction (Leifer and Huber, 1977).

TABLE 2

Correlations among determinants of boundary-spanning activity ($n=52$)*

	T	H	SS	TR	TI	PUT	PA	BS	BSA	BSB
T	-	0.401 ^c	0.054	-0.377 ^c	0.181	0.039	0.132	0.281 ^b	0.171	0.248 ^a
H		-	0.043	0.007	0.287 ^b	0.273 ^b	0.114	0.462 ^d	0.242 ^a	0.509 ^d
SS			-	-0.240 ^a	-0.006	0.330 ^b	-0.048	0.250 ^a	0.198	0.249 ^a
TR				-	-0.105	-0.132	0.081	-0.053	-0.030	-0.060
TI					-	0.248 ^a	0.104	-0.045	-0.221	0.024
PTU						-	0.048	0.105	-0.014	0.231 ^a
PA							-	-0.119	-0.158	-0.030
BS								-	0.847 ^d	0.921 ^d
BSA									-	0.687 ^d
BSB										-

Two-tailed test: * $p \leq 0.10$.

^b $p \leq 0.05$.

^c $p \leq 0.01$.

^d $p \leq 0.001$.

3. How are individual length of tenure with the EHV Program and level of hierarchy in project organization related to the perception of task uncertainty and level of boundary-spanning communication?

There is a significant association between tenure with the EHV Program and level in hierarchy ($r=0.401, p \leq 0.01$). Tenure with the EHV program shows positive associations with boundary-spanning overall ($r=0.281, p \leq 0.05$) and center-to-center ($r=0.248, p \leq 0.10$); while level in hierarchy is associated with perceived task uncertainty ($r=0.273, p \leq 0.05$) and overall boundary-spanning ($r=0.462, p=0.001$). The strongest association is between level of hierarchy and center-to-center boundary-spanning ($r=0.509; p \leq 0.001$). A further analysis controlling for tenure shows partial correlations between level in hierarchy and overall boundary-spanning ($0.273, p \leq 0.05$), center-to-center boundary-spanning ($0.419, p \leq 0.005$) and perceived task uncertainty ($0.342, p \leq 0.05$). These results show that (1) the longer the individual tenure with the EHV Program, the greater the level of hierarchy in project organization; and (2) controlling for tenure, the greater the level of hierarchy, the greater the perceived task uncertainty and the greater the level of center-to-center boundary-spanning

communication. These results are consistent with the findings of Bacharach and Aiken (1977) suggesting that a higher position in the organizational hierarchy individuals perceive greater levels of task uncertainty and engage in more communication activity across organizational boundaries.

4. How does perceived task uncertainty mediate the relationships between task-environmental characteristics and level of boundary-spanning communication?

Table 3 shows summaries of regression analyses for boundary-spanning communication. Only hierarchy in project organization and individual sensation-seeking tendency are significant antecedents of boundary-spanning communication. These two variables explain 28% (with correction for degrees of freedom suggested by Cohen and Cohen, 1975) of the variance in center-to-center boundary-spanning communication. The analyses revealed no evidence for a mediating effect of perceived task uncertainty on the relationships between task-environmental variables and level of boundary-spanning communication. The lack of evidence to substantiate the significance of perceived task uncertainty as a variable mediating the relationship between task-environmental char-

TABLE 3

Regression summary: The antecedents of boundary-spanning activity across project tasks within centers (A) and from center-to-center (B), and overall boundary-spanning ($n=52$)

Dependent variable	Independent variables in equation	Unstand. Reg. coef.	Stand. error	Stand. Part. Reg. coef.	<i>F</i>	Sig.	Mult. <i>R</i>	$\bar{R}^2$ ^a	Overall	
									<i>F</i>	Sig.
BSA	H	2.041	1.188	0.234	2.951	0.092	0.242	0.021	2.530	0.090
	SS	0.001	0.582	0.186	1.898	0.175	0.306	0.057		
	(Constant)	-1.305								
BSB	H	3.650	0.869	0.499	17.643	0.000	0.509	0.229	11.030	0.000
	SS	0.816	0.425	0.228	3.683	0.061	0.557	0.282		
	(Constant)	-4.531								
BS	H	3.27	0.886	0.452	13.639	0.001	0.462	0.182	8.903	0.009
	SS	0.815	0.433	0.230	3.537	0.066	0.516	0.237		
	(Constant)	-3.540								

^aCorrected for degrees of freedom (Cohen and Cohen, 1975).

acteristics and patterns of boundary-spanning communication is consistent with previous research (Bacharach and Aiken, 1977; Keller and Holland, 1975; Leifer and Huber, 1977; Tushman, 1977, 1979a, b).

RECOMMENDATIONS

These analyses have provided some preliminary answers to the research questions posed. Based on the results of this study in the perspective of the literature reviewed, the following recommendations for future research and theory development are formulated:

(1) Further exploration of relevant individual differences as antecedents of boundary-spanning communication, both within the project structure and across project boundaries is needed. Sensation-seeking tendency or other measures of risk pretence are of particular relevance as antecedents of arousal-seeking behavior (Mehrabian and Russell, 1974). demographic factors such as age and education should also be considered.

(2) Objectively measured organizational dimensions such as hierarchy, size, shape and

technology, having an impact on individual boundary-spanning activity need to be identified. Particular attention needs to be paid to this linkage between dimensions of organizational structure and behavioral outcomes at the individual level of analysis.

(3) Development of the "colleague role" concept suggested by Farris (1972) in the understanding of research and development communication patterns should be explored and developed. Technical information, organization information, and generation of original ideas are roles contributing to the "suggestion" phase in research and development problem-solving. The formulation of a proposal from a suggestion is facilitated by the colleague roles of help in thinking through the idea and critical evaluation of the idea. The proposal may be implemented through the contributing colleague roles of assuring a fair hearing and providing administrative assistance. Performance of the role of original idea generation (Farris, 1972) may be related to center-to-center boundary-spanning activity by individuals in positions at upper levels of hierarchy. Other roles suggested by Farris, such as help in thinking through an idea and critical evaluation of a proposal, may

be more strongly associated with intra-project boundary-spanning. A comprehensive approach to the concept of the "colleague role" may aid considerably in understanding patterns of boundary-spanning communication.

(4) A systematic approach to the development of measures of perceived task uncertainty should be pursued to determine the dimensionality of this important construct. Van de Ven (1977) defines the dimensions of perceived task uncertainty as task difficulty and variability consistent with Perrow (1967), without any empirical investigation of the structure of the measure. Such investigation could contribute to a better understanding of the construct and its significance to organizational behavior including communication patterns.

It is the hope of this researcher that these suggestions will provide an impetus to further research on the important topic of communication patterns in research and development settings and a better understanding of the management of innovative processes.

REFERENCES

- Bacharach, S.B. and Aiken, M., 1977. Communication in administrative bureaucracies. *Acad. Manage. J.*, 20(3): 365-377.
- Burns, T. and Stalker, G.M., 1961. *The Management of Innovation*. Tavistock, London.
- Campbell, J.P., Dunnette, M.D., Lawler, III, E.E. and Weick, Jr., K.E., 1970. *Managerial Behavior Performance and Effectiveness*. McGraw-Hill, New York, NY.
- Cohen, J. and Cohen, P., 1975. *Applied Multiple Regression/Correlational Analysis for the Behavioral Sciences*. John Wiley, New York, NY.
- Farris, G., 1972. The effect of individual roles on performance in innovative groups. *R&D Manage.*, 3: 23-28.
- Galbraith, J.R., 1977. *Organization Design*. Addison-Wesley, Reading, MA.
- House, R.J. and Rizzo, J.R., 1972. Toward the measurement of organizational practices: scale development and validation. *J. Appl. Psychol.*, 56: 388-396.
- Huber, G.P., O'Connell, M.J. and Cummings, L.L., 1975. Perceived environmental uncertainty: effects of information and structure. *Acad. Manage. J.*, 18: 725-740.
- James, R.J. and Jones, A.P., 1976. Organization structure: a review of structural dimensions and their conceptual relationship with individual attitudes and behavior. *Organ. Behav. Human. Perform.*, 16: 74-113.
- Keller, R. and Holland, W., 1975. Boundary spanning roles in an R&D organization. *Acad. Manage. J.*, 18: 388-393.
- Leifer, R. and Huber, G.P., 1977. Relations among perceived environmental uncertainty, organizational structure and boundary-spanning behavior. *Admin. Sci. Q.*, 22: 235-248.
- Mehrabian, A. and Russell, J.A., 1974. *An Approach to Environmental Psychology*. Cambridge, MA.
- Perrow, C., 1967. A framework for the comparative analysis of organizations. *Am. Sociol. Rev.*, 194-208.
- Rushman, M., 1977. Communication across organizational boundaries: special boundary roles in the innovation process. *Admin. Sci. Q.*, 22: 587-605.
- Tushman, M., 1979a. Determinants of subunit communication structure. *Admin. Sci. Q.*, 24: 82-98.
- Tushman, M., 1979b. Impacts of perceived environmental variability on patterns of work related communication. *Acad. Manage. J.*, 22: 482-500.
- Van de Ven, A.H., 1977. A panel study on the effects of task uncertainty, interdependence and size on unit decision making. *Organ. Admin. Sci.*, 8(2-3): 195-310.
- Van de Ven, A.H. and Delbecq, A., 1974. A task contingent model of work unit structure. *Admin. Sci. Q.*, 19: 183-197.
- Van de Ven, A.H., Delbecq, A. and Koenig, Jr., R., 1976. Determinants of coordination modes within organizations. *Am. Sociol. Rev.*, 41: 322-338.

APPENDIX A – INTERVIEW FORMAT

1. Task routinization

The following categories distinguish different types of research and development activities. Please read each definition carefully.

I. *Basic research*: work of a general nature intended to apply to a broad range of applications or to the development of new knowledge about an area.

II. *Applied research*: Work involving basic knowledge for the solution of a particular problem. The creation and evaluation of new concepts or components but not development for operational use.

III. *Development*: The combination of existing or feasible concepts, possibly with new knowledge, to provide a distinctly new product or process.

IV. *Technical services*: Cost/performance improvement to existing products, processes or systems. Recombination, modification and testing of systems using existing knowledge.

V. *Commercialization*: Opening new markets for existing products.

Roughly what percentage of the work being performed to accomplish your project's objectives falls in each of the five categories?

I: II: III: IV: V:

2. Planning activity

Please respond to the following statements with reference to planning activity for your *project*. There are two parts to each response: (a) Please rate the extent of your agreement or disagreement with each statement, and (b) Please rate the desirability of the condition described in each statement if it were true in your work environment.

(1) Important factors are frequently overlooked when plans are made.

- a. Definitely do not agree 1 2 3 4 5 6 7 Definitely agree
b. Extremely undesirable 1 2 3 4 5 6 7 extremely desirable

(2) Jobs are planned before they are started.

- a. Definitely do not agree 1 2 3 4 5 6 7 Definitely agree
b. Extremely undesirable 1 2 3 4 5 6 7 Extremely desirable

(3) Important details have not been considered when planning jobs.

- a. Definitely do not agree 1 2 3 4 5 6 7 Definitely agree
b. Extremely undesirable 1 2 3 4 5 6 7 Extremely desirable

(4) Work time is lost through poor scheduling and planning.

- a. Definitely do not agree 1 2 3 4 5 6 7 Definitely agree
b. Extremely undesirable 1 2 3 4 5 6 7 Extremely desirable

(5) When jobs are assigned, plans have been made to have all the necessary materials on hand.

- a. Definitely do not agree 1 2 3 4 5 6 7 Definitely agree
b. Extremely undesirable 1 2 3 4 5 6 7 Extremely desirable

3. Sensation-seeking tendency

Please indicate the extent of your agreement or disagreement with the following statements as descriptive of yourself.

- (1) I would like the job of foreign correspondent for a newspaper.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (2) I prefer an unpredictable life that is full of change to a more routine one
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (3) I like meeting people who give me new ideas.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (4) I would be content to live in the same town for the rest of my life.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (5) I like continually changing activities.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (6) I like a job that offers change, variety, and travel, even if it involves some danger.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (7) I like to look at pictures that are puzzling in some way.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (8) I much prefer familiar people and places.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (9) I like to touch and feel a sculpture.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (10) I don't enjoy doing daring, foolhardy things just for fun.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (11) I prefer a routine way of life to an unpredictable one full of change.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (12) I like to go somewhere different nearly every day.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (13) I am interested in new and varied interpretations of different art forms.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (14) I seldom change the pictures on my walls.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (15) I am continually seeking new ideas and experiences.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (16) I like to experience novelty and change in my daily routine.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement
- (17) I sometimes like to do things that are a little frightening.
Very strong disagreement 1 2 3 4 5 6 7 8 9 Very strong agreement

4. Task interdependence

Please indicate the degree to which your work is interrelated with the work of:

Technical personnel in other *laboratories*.

Very low interrelationship 1 2 3 4 5 6 7 8 9 10 Very high interrelationship

5. Perceived task uncertainty

Please respond to the following items as they describe your work on the EHV Program:

(1) To what extent is there a clearly defined body of knowledge or subject matter which can guide you in doing your work?

Not at all 1 2 3 4 5 6 7 8 9 10 To a very great extent

(2) To what extent is there an understandable sequence of steps to be followed in doing your work?

Not at all 1 2 3 4 5 6 7 8 9 10 To a very great extent

(3) During the course of your work, how often do you come across specific, difficult problems that you don't know how to solve immediately?

Never 1 2 3 4 5 6 7 8 9 10 Extremely frequently

(4) If there is something you don't know how to handle in your work, to what extent can you go to someone else for an answer to the problem?

Not at all 1 2 3 4 5 6 7 8 9 10 To a very great extent

(5) In some jobs, things are fairly predictable. In others, you are often not sure what the results of your efforts will be. What percent of the time are you generally sure what the outcome of your efforts will be?*

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

(6) On the average, how long is it before you know whether your effort is successful?*

Very immediate 1 2 3 4 5 6 7 8 9 10 Never

(7) How much variety do you generally encounter in your working day?

None 1 2 3 4 5 6 7 8 9 10 A very great amount

(8) To what extent are the activities or methods you follow in your work about the same dealing with various categories of problems?*

Not at all 1 2 3 4 5 6 7 8 9 10 To a very great extent

(9) To what extent would you say your work is routine?

Not at all 1 2 3 4 5 6 7 8 9 10 To a very great extent

Please indicate the extent to which each of the following statements is true or false:

(1) I do about the same job in the same way most of the time.

Very false 1 2 3 4 5 6 7 8 9 10 Very true

(2) Basically, I perform repetitive activities in doing my work.

Very false 1 2 3 4 5 6 7 8 9 10 Very true

*Items excluded from final format.

(3) In doing my job from day to day, I generally have to adopt different methods or procedures to do my work.

Very false 1 2 3 4 5 6 7 8 9 10 Very true

(4) There are different types or kinds of work to do every day on this project.

Very false 1 2 3 4 5 6 7 8 9 10 Very true

6. Boundary-spanning communication

In work related to the EHV Program during an average week.....

(1) How many hours do you attend formal meetings? (Committees, planning, etc.) (If not applicable, check here .)

(a) Outside your department but within the larger organization.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(b) Outside your organization altogether.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(2) How many hours do you confer about your work in informal face-to-face conversation or telephone conversation with people outside your immediate department? (If not applicable, check here .)

(a) Outside your department but within the larger organization.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(b) Outside your organization altogether.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(3) How many times do you send or receive formal, written but work related communications in the form of reports or data from people outside your immediate department? (If not applicable, check here .)

(a) Outside your department but within the larger organization.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(b) Outside your organization altogether.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(4) How many times do you send or receive informal, written communications in the form of reports, memos or the like from people outside your immediate department? (If not applicable, check here .)

(a) Outside your department but within the larger organization.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?

(b) Outside your organization altogether.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 How many more?